Specification
W138 Cavity Wax North
Cell 3 Robots Replacement

Version

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1. Update Service

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2. Update Journal

No.	Chapter	P.	Change reason	Contents of modification	Version	Date	Person responsible
01	All	All	New Document	Initial Version	01.00	Nov 2025	M. Terry

The current changes are yellow marked in relation to the older expenditure conditions

No. = Serial number **P.** = Page number **V.** = Version of change (document version)

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4. Foreword

All of the available **G**uide**l**ine's **GL-DE** (**De**utsch – German) and **GL-EN** (**En**glish) and **i**ntegra **M**aterial **R**elease **M**anager (**iMRM**) or **DEEP** (material approval lists) taken together provide the basis for consistent standardization of all technical products and services for manufacturing at Mercedes-Benz U.S. International, Inc. (MBUSI).

Details that are not precisely specified in these guidelines, or also items that are not covered in the associated electrical or mechanical standard documentation and in the appropriate software standards, shall be clarified with the person responsible at Mercedes-Benz AG.

The supplier shall develop proposals based on the existing guidelines, standard hardware documentation templates, and software standards and present them to the person responsible at Mercedes-Benz AG for approval.

The supplier shall inform Mercedes-Benz AG of any recommended improvements, modifications and additions that may be integrated into a subsequent version of these guidelines by Mercedes-Benz AG when they are updated.

This guideline does not release the supplier from its responsibility regarding the proper realization of the systems and documentation to be provided taking into account the valid and relevant laws, regulations, conditions and requirements at the respective production location.

If there are contradictions in the guidelines or invitations to tender, in case of doubt, the more stringent requirement shall apply.

ALL projects will follow EPG055 (MBUSI General Project Specification) and use MBUSI buyoff procedures. For all exceptions, a deviation request (EFG039) will have to be approved by the Standards Documentation lead and E4 manager. Approved MBUSI Buyoff Procedures:

- EPG030 (small projects, construction)
- EPG042 (production machinery with a new PLC being added to the plant

Please consult with the Standards and Controls Standards Documentation Lead for selection.

After project awarding, a kickoff meeting shall be held with the Standards and Controls Department to come to an agreement on documentation requirements for the project. The Documentation signature sheet will be sent to the Standards Documentation Lead to begin project tracking. Refer to EPG030/EPG042

A design meeting must occur before any project design is approved to verify design and parts compatibility with MBUSI parts obsolescence planning and all applicable standards, including MBUSI Plant Standards.

Signatures will be recorded on EFG079 signature sheets and emailed to the Standards Documentation Lead immediately after each milestone's signatures have been collected.
If an equivalent electronic system of any type is utilized in place of EFG079 (this requires MBUSI E4 approvals using form EFG039), then a digital or scanned copy from said system will be sent to the Standards Documentation Lead immediately after each milestone's signatures have been signed.

Standards Documentation Lead: Ann White**:** **ann.white@mercedes-benz.com**

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5. Basic Requirements for ECM-Based Project

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The total request consists of 3 volumes. These are located in the "Integra basic pool standard templates" project on an ECM server (Enterprise Content Management). For details, consult the following project description.

- Volume 1 - Integra Standard (non-project-specific part)
- Volume 2 - Part specific to project and machinery & equipment
- Volume 3 - Integra Standards

On submitting the tender the vendor acknowledges the terms of the RFQ (Volumes 1, 2 and 3) and undertakes to download these terms after setting ECM access and to comply with these terms in preparing the tender and in executing the order if subsequently placed. All deviations from the MB AG RFQ documents, Volumes 1 to 3, shall be disclosed in the corresponding chapter in Volume 2. The structure of the general chapters of Volumes 1 and 2 is identical.

Tenders can only be evaluated and considered in contract award if they have been compiled in a legally binding manner on the forms specified by MB AG, signed, and submitted on time.

The requirements formulated in this project-specific part of the RFQ do not exempt the supplier from its responsibility for preparing the annexes and documents to be supplied by it in a professional manner and in compliance with the laws, regulations, and requirements in force at the production site in question.

Access to the Customer Specification Documents
The objective of the procedure described in this chapter is to exchange the RFQ and project documents between the vendor or contractor and Mercedes-Benz AG as the client without paper and automatically as information in file form via the Internet.

System Prerequisites / Access Authorization
All RFQ documents required for preparing the tender are located on an internet server (ECM server). For downloading this content, the vendor or contractor needs special access authorization to be set up before preparing the tender.

The vendor will receive for this purpose together with Volume 2 instructions on how to have an ECM user account created and access the RFQ documents of the Standard Integra through the internet. To this end, the responsible MB AG project manager shall e-mail a request form (Excel file) to the vendor for recording the data of the new ECM user to be created. For the bidder to have access to the relevant customer specification documents, the bidder is requested to complete this request form in advance and return it to the responsible MB AG project manager. The request form e-mailed back is used to create the new user account in the system. At the same time, the vendor acknowledges the non-disclosure agreement in this request form.
The process of creating a user occurs just once. In the event of further projects with the Mercedes Car Group this ECM access will continue to be available to the vendor.
Should there be any questions about setting up the access to ECM, the vendor can use the following e-mail address: ecm-pp@mercedes-benz.com

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On submitting the tender, the vendor acknowledges the terms of the RFQ and to comply with these terms in preparing the tender and in executing the order if subsequently placed.
All deviations from the MB AG RFQ documents, shall be disclosed.
Tenders can only be evaluated and considered in contract award if they have been compiled in a legally binding manner on the forms specified by MB AG, signed and submitted on time.
The requirements formulated in this project-specific part of the RFQ do not exempt the supplier from its responsibility for preparing the annexes and documents to be supplied by it in a professional manner and in compliance with the laws, regulations and requirements in force at the production site in question.

7. Project Specifics

The existing Cavity Wax North Cell 3 robots have reached the end of their operational lifespan, leading to increased maintenance costs and reduced reliability. Upgrading these robots is essential for maintaining production efficiency and minimizing downtime.

This project aims to replace four (4) ABB IRB 4400L Cavity Wax process robots in North Cell 3 due to equipment age and obsolescence. The project includes upgrading associated components to enhance operational efficiency and reliability. This is a turnkey project, and it is the supplier's responsibility to meet all project objectives. These upgrades will include the cavity wax robot's process equipment and 7th axis rail. The ABB IRB 4400L robots are no longer provided by ABB and the supplier will provide the appropriate replacement robot to meet all project objectives including all required engineering validation, simulation, and performance verification. The project installation is scheduled for winter shutdown 2026.

Project Objectives

The project's objectives are to meet production targets and stringent quality requirements during the installation and commissioning phases in the Paint Shop while also optimizing efficiency and minimizing disruption:

1. Meeting production and cycle time targets: The project aims for a cycle time that is equal to or better than the baseline at project kick-off.
2. Quality Standards: The result meets or exceeds all specified quality requirements as defined by the MBUSI q-release process.
3. Zero Unscheduled Downtime: The project is planned to avoid any unplanned interruptions, ensuring that normal production activities continue without hindrance.
4. Accurate and Updated Documentation: All relevant documentation, including buyoff tags, electrical, mechanical, and controls drawings and layouts, will be updated to reflect the final as-built status of the project.
5. A successful buyoff with signatures from all relevant stakeholders at each project phase. MBUSI buyoff procedure EPG030 will be used.

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8. Tasks

Under the turn-key concept, the supplier shall assume full and complete responsibility for all aspects of the project. The supplier is responsible for furnishing all hardware, equipment, materials, and services required for the successful completion of the work. This includes retaining, supervising, and managing all electrical and mechanical subcontractors, ensuring that only MBUSI-approved contractors are used.

The supplier is fully responsible for compliance with all MBUSI electrical, mechanical, safety, and environmental specifications, as well as all shop-specific rules, policies, and procedures. The supplier is also responsible for planning, coordinating, executing, and validating all work necessary to achieve successful start-up and production readiness.

Any upgrades, modifications, interfaces, or adjustments required for system integration shall be the sole responsibility of the supplier.

This turn-key installation/retrofit includes all required design, engineering, fabrication, procurement, installation, demolition, rework, testing, commissioning, documentation, and training necessary to meet all project objectives and deliver a complete and functional system.

This is a turn-key project. The supplier assumes full responsibility for the entire scope and all outcomes. Change orders will be tightly controlled and kept to a minimum.

Mercedes-Benz Enterprise Content Management (ECM)
Refer to ECM for as-built documentation package, plant specific standards, and Integra References, and W138 Rockwell data pool.

9. Cavity Wax North Cell 3 Robots

The four (4) ABB IRB 4400L/30 process robots currently installed in Cell 3 must be replaced with new robots that are functionally equivalent in terms of performance, reach, and payload capacity. The replacement robots shall be fully integrated with fluid control, Festo valve systems, PLC controls, Beckhoff HMI screens, and plant SQL reporting consistent with Cavity Wax North Cells 1 & 2.

The replacement robots must be capable of performing all existing process operations without requiring significant modifications to the cell layout, tooling, or interfaces. In addition to the robots, the associated controllers must also be replaced to ensure full compatibility with the new equipment. The new robot controllers should integrate seamlessly with existing plant networks and safety systems while maintaining or improving overall system reliability and performance. The awarded supplier is fully responsible for selection and validation of the replacement robot platform including payload, reach, dress package weight, and dynamic performance. The supplier shall perform and submit a full robot simulation and payload validation demonstrating that the selected robot can perform all existing processes with a minimum 20% payload safety margin and without degradation in cycle time, accuracy, or accessibility.

All modifications required due to robot selection (dress pack, EOAT, rail position, mounting, or layout adjustments) are included in the supplier scope at no additional cost.

The supplier is responsible for providing robot controllers compatible with the selected robot platform and compliant with ABB current production standards.

MBUSI has four (4) new IRC5 controllers that can be used. If the supplier elects to use these

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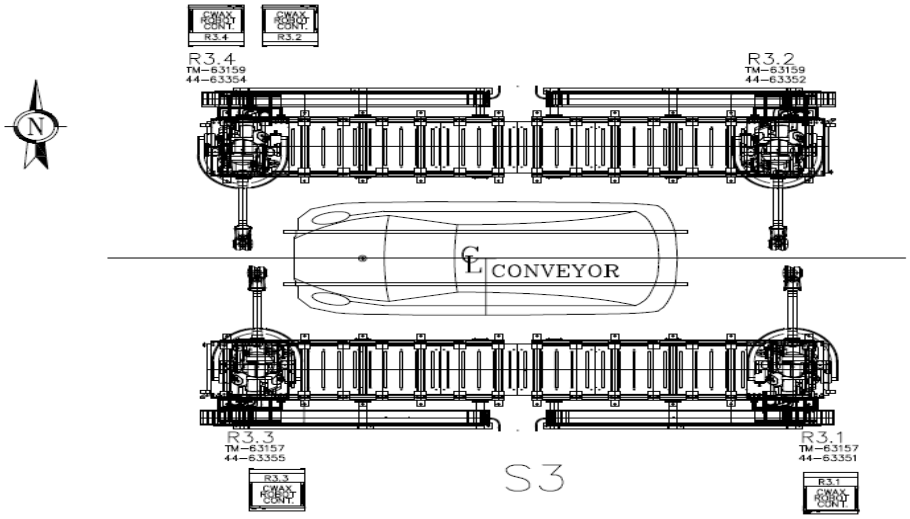
existing IRC5 controllers, the supplier is responsible for verifying compatibility, performing all required hardware and software modifications, and guaranteeing full functionality, performance, and long-term supportability. Any additional drive units, communication hardware, safety modules, or software upgrades required to integrate existing IRC5 controllers are included in the supplier scope.

Serial numbers for the new IRC5 controllers that MBUSI can provide if the supplier chooses to use are shown below:

IRC5 controller material number: 3HAC020536-015

Serial Numbers:
05-800586
05-800583
05-800589
05-800584



10. 7th Axis Rails

The 7th axis robot rail assemblies associated with the four (4) process robots in Cell 3 must be fully replaced as part of this project scope. The new rails must match the existing system in length, travel distance, mounting configuration, and load capacity to ensure compatibility with the replacement robots and associated tooling.

All rail components, including the drive system, motor, encoder, gear unit, and cable management assemblies, must be equivalent in function and performance to the current design. The new system must allow for identical robot positioning, motion range, and speed as the existing installation to maintain established process timing and accuracy.

If any existing rail components or assemblies cannot be sourced, equivalent replacements may be proposed but must receive prior approval from MBUSI. The replacement rails shall be fully installed, aligned, and functionally tested to verify smooth operation, repeatability, and integration with the

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new robot controllers and safety systems.

The supplier shall replace the 7th axis rail lubrication system to current MBUSI standard.

11. Process Equipment

All process equipment from the robot base through the end-of-arm tooling must be replaced as part of this scope. The replacement equipment must exactly match the specifications, configuration, and functionality of the existing components, including the Leoni hose packaging and routing. Any deviation from the current design, materials, or component specifications is not permitted unless the original equipment can no longer be sourced. In such cases, equivalent replacement parts or assemblies may be proposed but must receive prior approval from MBUSI before procurement or installation. All replacement equipment shall be installed and validated to ensure proper integration, performance, and durability consistent with existing production standards.

Four (4) new nozzle heads must be purchased and included in the modular build. IPR parts numbers are shown below:

152108802 DWK 4 Way
15210505 Coding DWK 4-Wa

Fluid System, Measurement, and Process Controls

The supplier shall provide a complete fluid delivery, measurement, and control system fully integrated with the robot, PLC, and Beckhoff HMI platform. The system shall be based on the current standard implemented in Cavity Wax North Cells 1 & 2 and shall match or exceed the performance and functionality of those systems.

The scope shall include, but is not limited to, the following:

Fluid Control and Measurement

- All flow meters shall be capable of beaker verification testing.
  The system shall include all hardware, valving, and programming required to allow operators and maintenance to perform manual beaker tests to verify flow meter accuracy.
- The system shall include low flow detection and control functionality equivalent to Cells 1 & 2.
- Flow meters shall be calibrated and validated during SAT with documented accuracy results.
- All flow, pressure, and volume parameters shall be displayed on Beckhoff HMI screens

Volume Tracking and Data Storage

- The system shall record the applied volume for each nozzle hole / head per vehicle.
- All process data shall be stored in MBUSI SQL reporting systems consistent with existing wax cells.
- Data structure, tags, and reporting shall match the standard used in Cells 1 & 2 to ensure data consistency.

Fluid Control Screens (HMI)

- Beckhoff HMI screens shall be provided and shall function identically to Cells 1 & 2, including:
  - Flow rate monitoring
  - Pressure monitoring
  - Low flow alarms
  - Beaker test interface

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- Volume tracking

Dress Pack and Fluid Routing

- The robot dress package shall include all fluid lines, sensor wiring, and protection required for long-term durability.
- The design shall match the current MBUSI standard dress package used in Cells 1 & 2.

End of Arm Tooling (EOAT) / Head Interface

- Supplier shall provide new EOAT receiver interfaces for robot head pickup
- All EOAT mechanical and electrical interfaces shall be validated for repeatability and reliability.

Festo Valve Pack

- The supplier shall replace the existing SMC box with a Festo valve pack mounted on top of the robot, matching exisiting MBUSI standard architecture.
- All pneumatic control shall be integrated through this Festo system.

12. Substructure and Vibration Isolation

All automation equipment, particularly the robots, shall be installed on a vibration-free and structurally independent substructure, fully decoupled from the booth design.

The robot supplier shall provide a steel substructure capable of supporting both static and dynamic loads associated with robot operation. This structure must maintain rigidity such that total deflection due to dynamic motion, both internal and external, does not exceed ±1.0 mm at the robot tool center point (TCP) throughout the entire working area.

All necessary adjustments, including machined plates, leveling shims, and alignment procedures, are included within the automation supplier's scope of supply. The supplier shall also identify and document all static and dynamic loads in their proposal. Measurement of structural performance shall be conducted under maximum acceleration, in accordance with the paint automation specifications. To prevent vibration amplification caused by interference between the booth and substructure, both systems shall be completely decoupled.

13. Booth Reference Monument and TCP

As part of the robot integration, Tool Center Points (TCPs) must be precisely defined, validated, and documented. The TCPs and their validation methods shall match the existing configuration and verification standards used in the Cavity Wax South Cells to maintain process consistency and repeatability. A reference monument shall be installed at the center of the application area, positioned so that all four robots have full access. The layout and monument configuration shall mirror the existing Cavity Wax South Cell design to ensure alignment between systems. The designated test point on the monument must be accessible in semi-automatic mode for verification purposes. Upon completion of commissioning, the supplier shall prepare and deliver a test certificate verifying TCP accuracy and repeatability to MBUSI.

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14. Surveys and Services

Before commencing work, this Contractor shall submit a list or survey indicating all existing damages and deficiencies, which are to remain. This list or survey will serve as a benchmark during punch-list to verify additional damage done during construction.
This Contractor shall be held liable to repair all damages caused by their work or their Sub-Contractors, not previously identified, at no cost to the Owner.
The Contractor will be held responsible for any floor damage caused by his workmen and shall make repairs to the damaged area to the satisfaction of and at no cost to the Owner. All anchor bolts, fasteners, conduits, pipe, rails, etc. which are embedded in the concrete floor shall be cut and ground smooth, flush with finish floor.

15. Requirements and Design

Wax Cell 3 Takt Time 108s, 18s transfer and 90s process. This is for all models.
Performance

Annual production:	333.000 u/a
Technical availability of the system:	P = 99.5 %

All design and installation activities shall strictly comply with the ECM. No deviations from ECM standards are permitted without prior written approval from the MBUSI Project Leader. The supplier shall ensure that all system additions include every component necessary to support both automatic and manual operation, including any required visualization updates to the HMI to maintain full system functionality and operator visibility. A full digital simulation including robot reach, cycle time, interference check, and process coverage shall be submitted and approved by MBUSI prior to equipment procurement and FAT.

Controls, PLC, Beckhoff HMI, and Communications Integration

The supplier shall provide a complete controls and communications architecture fully integrated into MBUSI Paint Shop standards.

PLC / Controls Integration

- The system shall interface with the existing Paint Shop PLC architecture and follow Integra programming standards.
- All robot, fluid, Festo, and safety I/O shall be integrated into the plant PLC system.

Beckhoff HMI Integration

- All interfaces, feedback signals, and diagnostics shall be fully available on the HMI.
- HMI functionality shall match Cells 1 & 2.

Robot Communication

- Ethernet communication shall be provided between:
  - Robot controllers
  - PLC
  - Beckhoff HMI
  - Process equipment

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ISRA Vision Integration

- The system shall accept offset data from ISRA vision systems and communicate offsets directly to the robot controllers.
- All communication protocols and data mapping shall be included in supplier scope.

Job Data / Head Selection

- The supplier shall implement logic for job data transmission to robots to ensure correct head selection and process recipe execution.

Safety Integration

- All safety interlocks from the existing PLC safety system shall be integrated into the robot controllers.
- The supplier shall provide all required safety programming, validation, and documentation.

The supplier shall provide all wiring, labeling, junction boxes, conduit, and mounting hardware necessary to fully meet MBUSI specifications. All safety hardware, including switches, relays, guards, and related components, must be installed in accordance with ECM and MBUSI Safety Standards. The supplier is responsible for all Safety PLC programming necessary to achieve a compliant and validated system design.

Any omissions, discrepancies, or conflicts identified in the design or documentation shall be reported to the MBUSI Project Team prior to implementing any changes. Rework or design modifications must ensure that existing process clearances and vehicle protective shielding are not altered. Any deviation that affects these parameters must receive explicit MBUSI approval before proceeding.

All materials and chemical products used during fabrication, installation, or operation must be selected from the MBUSI pre-approved Paint Shop Compatibility List and must have an existing MSC code. If no pre-approved product is available, the supplier must submit the proposed material for MBUSI approval prior to bringing it onsite and ensure an MSC code is generated. All containers must be clearly labeled with the corresponding MSC code.

All materials, tools, and equipment used within the Paint Shop must be 100% silicone-free. No materials containing silicone will be permitted under any circumstances.

16. Factory Acceptance Testing – Mod Build

A comprehensive Modular Build (Factory Acceptance Test) shall be conducted prior to shipment of any equipment to MBUSI. The Modular Build (FAT) shall include full demonstration of robot paths and precision. All shipping costs related to modular build are the responsibility of the supplier.

MBUSI Engineering and Maintenance personnel will participate in the Modular Build to confirm that the system meets MBUSI quality and performance standards. The buy-off process shall follow the current MBUSI Modular Build (FAT) procedures as agreed upon by the project team. Inspections, including motion verification, process validation, safety interlocks, and communication checks, must be documented and included in the final FAT report.

Any deviations from the approved Modular Build scope, procedures, or acceptance criteria must receive written approval from the designated MBUSI Project Leader prior to project start. The Modular Build and FAT execution are scheduled to occur in **November 2026**, with full participation from MBUSI Engineering, Maintenance, and Project Management representatives.

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17. Installation and Commissioning

Following successful completion and approval of the Modular Build (FAT), the complete system—including robots, controllers, process equipment, and 7th axis rails—shall be disassembled, shipped, and reinstalled at MBUSI. The installation contractor shall be responsible for coordinating all site work in accordance with MBUSI safety, quality, and environmental requirements.

The supplier shall perform on-site validation of flow meter accuracy using a beaker test method to confirm proper measurement and calibration. The supplier shall also verify SQL reporting functionality and confirm the integrity and accuracy of all recorded data. ISRA offsets shall be validated under actual production conditions to ensure proper system alignment and performance. The supplier shall complete full safety validation of all PLC-to-robot interlocks to ensure proper functionality and compliance with all applicable safety requirements.

The installation contractor must obtain all required MBUSI permits prior to performing any work, including but not limited to the Safe Work Permit and Hot Work Permit. All MBUSI safe work practices, such as Lock Out/Tag Out (LOTO), must be strictly followed throughout the duration of the project. The contractor shall ensure that all personnel are properly trained, qualified, and authorized to perform assigned tasks safely and in compliance with MBUSI standards.

The contractor is responsible for the removal, modification, and reinstallation of all hardware, software, and related equipment or devices required to complete the project within the defined work area. All equipment shall be reinstalled to match the configuration demonstrated during the Modular Build, including cable routing, hose management, and mechanical alignment. The work area is to be kept clean and left clean after installation is complete. The MBUSI Project Leader or their designee must review and approve the area once all modifications and cleaning have been completed.

All programming work shall be performed by MBUSI-approved software integration engineers. The contractor will also be responsible for the successful startup and debug of all modified zones' control systems. The principal contractor must be present on site to supervise all electrical installation work. Programming for the Paint Shop must follow the Integra standard for conveyor and process integration, and the supplier is responsible for obtaining and using the Paint Shop–specific Integra programming guidelines. All control logic for the Paint Shop shall be written in ladder format.

During commissioning, the supplier shall perform complete system power-up, input/output verification, and functional testing to confirm proper operation of all components. Testing shall include validation of robot motion paths, wax application accuracy, communication with plant systems, and safety interlock functionality. All gauges must be marked with acceptable running ranges, all fasteners must be witness-marked, and all wires, cables, and hoses must be labeled consistent with existing MBUSI standards.

Commissioning will be witnessed by MBUSI Engineering and Maintenance representatives to verify that the system meets all operational, safety, and quality requirements. A Site Acceptance Test (SAT) will be conducted to confirm readiness for production. Any deviations or corrective actions identified during commissioning or SAT must be documented and resolved before final project closeout. Upon completion, the contractor shall provide MBUSI with all related documentation, including parameter backups, program versions, maintenance manuals, and as-built drawings.

18. Start-up, Training, and Standby Support

Upon successful completion of the Site Acceptance Test (SAT) and resolution of all outstanding issues, the system shall be formally handed over to MBUSI. The contractor is responsible for the

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satisfactory start-up of all equipment provided under this contract. The start-up period will begin on the first production day following the shutdown period when commissioning of equipment occurs. During this phase, the contractor shall actively support shop moves and shop trials as part of the installation and commissioning process. This period is considered an essential phase of the project and shall not be regarded as standby support.

Before start-up support may conclude, a minimum of two (2) full production weeks must have passed, and Q-release must be achieved on all vehicle models. During this time, the contractor is responsible for maintaining a sufficient and qualified crew for each trade involved, ensuring prompt correction of any deficiencies that are their responsibility. The contractor shall also maintain an adequate inventory of spare parts for all new equipment from initial power-on through final acceptance, in accordance with the Master Equipment Agreement (MEA).

All as-built documentation—including electrical and mechanical drawings, PLC and robot program backups, parameter settings, and system configurations—shall be submitted in both digital and printed formats. Final versions of manuals, maintenance procedures, spare parts lists, and recovery instructions must accurately reflect the as-installed condition and must be approved by MBUSI Engineering prior to project closeout.

Comprehensive systems training shall be provided for all MBUSI Maintenance teams, covering equipment operation, troubleshooting, and recovery procedures. A detailed training outline shall be submitted to MBUSI for approval at least two (2) weeks prior to the start of training. Training may occur during the standby support period and must include both classroom and hands-on instruction using production equipment. Training materials shall be provided in English, and attendance records must be submitted to MBUSI upon completion.

The supplier and contractor shall provide continued technical support throughout the production ramp-up phase to assist with process optimization and system fine-tuning. Any issues identified during this period must be addressed promptly at no additional cost to MBUSI. Final project acceptance will be granted only after all documentation, training, and start-up support requirements have been fulfilled to MBUSI's satisfaction and all performance criteria have been achieved.

19. Material Logistics / Scrap Material

All scrap materials will be removed from the plant by the installation contractors and disposed of in gondolas provided by the supplier.

All lay-down areas for the project must have approval by MBUSI engineering.

For all Material transports, the route through the paint shop shall be approved by MBUSI engineering.

Material transportation through the building will be limited to non-production hours unless approved by MBUSI engineering. All materials must be cleaned of excess dirt prior to entering the building.

20. Deliverables/Responsibilities

The contractor shall submit a complete manpower list prior to the start of the project. This list must include all direct employees as well as any subcontractors who will participate in the work. All personnel are subject to approval by MBUSI before beginning any on-site activities.

The contractor is responsible for developing and maintaining a project schedule that aligns with MBUSI's fabrication, delivery, and installation timelines. This includes planning and executing all necessary overtime, weekend, and shift work required to meet deadlines without additional cost to MBUSI. The contractor shall report daily project progress to the Engineer in Charge (EIC) and

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promptly communicate any issues that may impact the schedule or project scope.

All work must conform to MBUSI's electrical, mechanical, and safe work specifications, as well as any shop-specific rules and policies. The contractor is also responsible for coordinating and supervising all subcontractors involved in the installation. It is the contractor's duty to ensure that all subcontractor equipment and installations comply fully with MBUSI standards and specifications.

Any changes in the project scope must be approved in writing by MBUSI prior to execution. The contractor will bear all costs associated with performing unapproved or verbally authorized changes. Final measurements and dimensions for all installation work are the sole responsibility of the installation contractor and must be verified for accuracy prior to completion.

The supplier is responsible for managing all equipment lead times, including robots, controllers, rails, and process equipment, to meet the MBUSI installation schedule.

Any expediting, premium freight, or schedule recovery actions required to meet project milestones shall be at the supplier's expense and are included in the base scope.

21. Project Milestones, Buy Off Procedure, and Documentation

The project shall target completion no later than the date specified in the approved project schedule. The contractor will develop and maintain a single, comprehensive project schedule in coordination with MBUSI. This schedule will be mutually developed, reviewed, and approved by both parties prior to project commencement. The official format for the schedule shall be a Microsoft Project Gantt chart.

The contractor is responsible for tracking and reporting progress against this schedule, including milestone achievements, daily progress updates, and implementation of countermeasures in the event of missed milestones. Project milestones will include, but are not limited to, personnel planning, parts delivery, mechanical installation completion, trial completion, and the projected buy-off date.

The official buy-off procedure for the project shall follow EPG030, which will serve as the governing standard for all buy-off activities and documentation. Milestones will be based on the stages defined within EPG030, ensuring alignment with MBUSI's established approval process. In addition to the EPG030 milestones, a shipping release milestone shall be completed during factory acceptance testing to confirm readiness for shipment.

All related documentation and approvals shall be managed and recorded within the DEEP process to ensure full traceability and compliance with MBUSI requirements. It is the responsibility of the supplier to ensure documentation meets DEEP requirements. MBUSI will provide DEEP training to one team member of the awarded supplier.

22. Alternative solutions, proposals and concepts

MBUSI expressly requests that the supplier submit alternative proposals in addition to the base scope of supply if any are available. These alternatives should present solutions that may offer improved performance, increased efficiency, or cost savings while maintaining adherence to the fundamental concept and functional requirements defined by MBUSI.

Each alternative proposal must include detailed descriptions and clear explanations outlining the technical and commercial advantages of the proposed deviation. Alternative quotations shall be submitted in
addition to, and not in place of, the quotation prepared in accordance with the official table of

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services and supplies. Submission of alternatives does not release the supplier from the obligation to provide a complete quotation fully compliant with the defined scope.

All alternative quotations must be prepared in the same format as the base quotation to allow for direct technical and commercial comparison. The specific items within the scope of supply for which the alternatives are being proposed must be clearly identified. The documentation submitted shall enable a straightforward technical comparison between the alternative proposal and the basic quotation to support MBUSI's evaluation and decision-making process.

The supplier's proposal shall include all costs required to meet the functional requirements of this specification. No additional costs related to robot selection, controller changes, payload limitations, reach limitations, or equipment obsolescence will be accepted after contract award.

23. Process Compliance

The installation contractor shall be responsible for chairing the weekly weekend work meeting held the weekend prior to scheduled work. During this meeting, the contractor shall review the upcoming work plan, manpower, safety measures, and potential impacts to production. If the contractor is unable to attend, MBUSI Engineering must be notified in advance, and detailed information—including dates, times, locations, and scope of work—shall be submitted prior to the meeting commencing.

Once the shutdown period has begun, the contractor shall ensure that a designated representative attends the daily morning meeting (time to be determined) to provide updates on progress, completed activities, and upcoming tasks. Any project issues, delays, or safety concerns shall be brought forward and addressed at this time in coordination with MBUSI Engineering and Maintenance.

Throughout the duration of the project, the contractor shall also chair weekly design meetings to review system design status, interface requirements, open issues, and any proposed modifications. These meetings shall include MBUSI Engineering and relevant supplier representatives to ensure timely resolution of technical matters and alignment with project specifications and schedule. The contractor is responsible for preparing meeting agendas, recording minutes, and distributing action items and follow-up responsibilities to all attendees.

24. Documentation Requirements

DEEP process is required. See ECM

25. References

AAW's for all five (5) models
Buy off Procedure EPG 030
Cavity Wax North Cells 1 & 2 – Controls, Fluid System, and Beckhoff HMI Standard Architecture
Cavity Wax North Cell 3 Refurbishment Timeline.pdf
ECM Project: BR267-P22P-W138- Cavity Wax North Cell 3 Robot Refurbishment
Volume 1 gives general instructions
Volume 2 includes the project specifications (RFQ)
Volume 3 is the first source for standards, if topics are not described in this folder, see Volume 3 (i6 and i7)

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26. Quotation Requirements

Quotation will be emailed to:
ramona.ludgood@mercedesbenz.com

Cost must be itemized for Labor, Material, and Equipment. Expenses must fall under MBUSI expense guidelines.

Initial RFQ is due: see Ramona Ludgood

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